

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277940

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Lee; Sanggeon et al.

OPTICAL FIBER ATTACHMENT DEVICE

Abstract

An optical fiber mold device has a first portion that includes a base layer having a longitudinal feature configured to receive an optical fiber. At least one second portion is disposed over the base layer. The second portion has a center wall and front and back end walls. The center wall, the front end wall, and the back end wall form a mold cavity. At least one first hole is disposed in the mold cavity and is configured to allow mold material to enter the mold cavity. At least one second hole in the mold cavity is configured to allow air displaced by the mold material to exit the mold cavity.

Inventors: Lee; Sanggeon (San Ramon, CA), Kiesel; Peter (Palo Alto, CA), Chen; Qiushu (San Jose, CA), Chamoun; Jacob N. (Somerville, MA)

Applicant: Xerox Corporation (Norwalk, CT)

Family ID: 82846271

Appl. No.: 19/210322

Filed: May 16, 2025

Related U.S. Application Data

parent US division 17409408 20210823 parent-grant-document US 12306443 child US 19210322

Publication Classification

Int. Cl.: G02B6/36 (20060101); B29C45/14 (20060101); B29C45/17 (20060101); B29C45/34 (20060101); B29D11/00 (20060101); B29K63/00 (20060101); B29L11/00 (20060101); C03B37/10 (20060101); G01L1/24 (20060101)

U.S. Cl.:

CPC G02B6/3648 (20130101); B29C45/14065 (20130101); B29C45/14467 (20130101); B29C45/14549 (20130101); B29C45/1742 (20130101); B29C45/34 (20130101);

Background/Summary

BACKGROUND

[0001] Fiber optic (FO) sensors can be used for detecting parameters such as strain, temperature, pressure, current, voltage, chemical composition, and vibration. FO sensors are attractive components because they are thin, lightweight, sensitive, robust to harsh environments, and immune to electromagnetic interference (EMI) and electrostatic discharge.

[0002] FO sensors can be arranged to simultaneously measure multiple parameters distributed in space with high sensitivity in multiplexed configurations over long optical fiber cables. One example of how this can be achieved is through fiber Bragg grating (FBG) sensors. A FBG sensor is formed by a periodic modulation of the refractive index along a finite length (typically a few mm) of the core of an optical fiber. This pattern reflects a wavelength, called the Bragg wavelength, determined by the periodicity of the refractive index profile. The Bragg wavelength is sensitive to external stimulus (strain and/or temperature, etc.) that changes the periodicity of the grating and/or the index of refraction of the fiber. Thus, FBG sensors rely on the detection of small wavelength changes in response to stimuli of interest. In some implementations, FO sensors can be attached to structures and operated to detect parameters, e.g., strain, temperature, vibration, related to the health of the structures.

SUMMARY

[0003] Embodiments described herein involve an optical fiber mold device, comprising a first portion that includes a base layer having a longitudinal feature configured to receive an optical fiber. At least one second portion is disposed over the base layer. The second portion comprises a center wall and front and back end walls. The center wall, the front end wall, and the back end wall forming a mold cavity. At least one first hole is disposed in the mold cavity and is configured to allow mold material to enter the mold cavity. At least one second hole in the mold cavity is configured to allow air displaced by the mold material to exit the mold cavity.

[0004] An optical fiber mold device comprises a first portion that includes a base layer having a longitudinal feature configured to receive an optical fiber. Two or more second portions are spaced apart on the base layer. Each second portion comprises a center wall and front and back end walls. The center wall, the front end wall, and the back end wall forming a mold cavity. At least one first hole is disposed in the mold cavity and is configured to allow mold material to enter the mold cavity. At least one second hole is disposed in the mold cavity and is configured to allow air displaced by the mold material to exit the mold cavity. A tunnel portion is disposed between the two second portions. The tunnel portion is configured to contain an at least one optical sensor disposed on the optical fiber.

[0005] A method involves positioning an optical fiber in a longitudinal feature of a base layer of a mold device. The mold device comprises a first portion comprising the base layer having the longitudinal feature configured to receive the optical fiber. A second portion disposed on the base layer. The second portion comprises a center wall and front and back end walls. The center wall, the front end wall, and the back end wall forming a mold cavity.

[0006] At least one first hole is disposed in the mold cavity and is configured to allow mold material to enter the mold cavity. At least one second hole is disposed in the mold cavity and is configured to allow air displaced by the mold material to exit the mold cavity. The method includes

placing the mold device and the optical fiber on a structure surface and injecting the mold material into the first hole of the mold device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Throughout the specification reference is made to the appended drawings wherein: FIG. 1A is a perspective view of a mold device in accordance with some embodiments;

[0008] FIG. 1B and 1C are bottom view of the mold device of FIG. 1A;

[0009] FIG. 1D and 1E are example cross sectional views of the mold device of FIG. 1A;

[0010] FIG. 2 is a perspective view of a mold device that includes a notch in the front and back walls in accordance with some embodiments;

[0011] FIG. 3 is a perspective view of a mold device having front and back extensions in accordance with some embodiments;

[0012] FIG. 4 is a top view of a mold device having curved front and back end walls in accordance with some embodiments;

[0013] FIG. 5A is a side cross sectional view of a mold device having adhesive disposed on the lower surface of the mold device in accordance with some embodiments;

[0014] FIG. 5B is a bottom view of the mold device of FIG. 5A;

[0015] FIG. 5C is a side cross sectional view of a mold device having adhesive disposed on the lower surfaces of the side extensions of the mold device in accordance with some embodiments;

[0016] FIG. 5D is a bottom view of the mold device of FIG. 5C;

[0017] FIG. 6A is a side cross sectional view of a mold device having adhesive disposed on the upper surfaces of the side extensions of the mold device in accordance with some embodiments;

[0018] FIG. 6B is a bottom view of the mold device of FIG. 6A;

[0019] FIG. 6C is a perspective view of the mold device of FIG. 6A;

[0020] FIG. 7 is a perspective view of a mold device having a flowable adhesive disposed on the upper surfaces of side extensions of the mold device in accordance with some embodiments;

[0021] FIG. 8A is a bottom view of a mold device having adhesive disposed on bottom surfaces of the side extensions and front and back extensions in accordance with some embodiments;

[0022] FIG. 8B is a side cross sectional view of the mold device of FIG. 8A;

[0023] FIG. 9A is a bottom view of a mold device having an adhesive tape disposed on the upper surfaces of the mold device in accordance with some embodiments;

[0024] FIG. 9B is a perspective view of the mold device of FIG. 9A;

[0025] FIG. 10 is a top view of a mold device having engagement feature disposed on the base layer in accordance with some embodiments;

[0026] FIG. 11A shows mold device arranged in a rosette pattern in accordance with some embodiments;

[0027] FIG. 11B shows representative dimensions for the mold devices of FIG. 11A;

[0028] FIG. 12 shows mold devices arranged in a biaxial pattern in accordance with some embodiments;

[0029] FIGS. 13A and 13B are flow diagrams illustrating processes of using mold devices in accordance with some embodiments;

[0030] FIG. 14 illustrates a process of attaching mold devices in accordance with some embodiments;

[0031] FIG. 15A through 15Q illustrate a process of attaching mold devices to a structure, the mold devices having front and back features used to pre-strain the optical fibers in accordance with some embodiments;

[0032] FIG. 16 is a top view of a mold device useful for attaching an optical fiber and/or sensor

such that the sensor is substantially strain-isolated from the structure in accordance with some embodiments; and

[0033] FIGS. **17A** and **17B** show views of a mold device suitable for attaching an optical fiber comprising a strain free temperature sensor and/or a span sensor in accordance with some embodiments.

[0034] The figures are not necessarily to scale. Like numbers used in the figures refer to like components. However, it will be understood that the use of a number to refer to a component in a given figure is not intended to limit the component in another figure labeled with the same number.

DETAILED DESCRIPTION

[0035] Fiber optic (FO) sensors can be deployed on various types of structures, e.g., bridges, roadways, railways, and electrical devices such as transformers, to monitor the structural health of the structures. Embodiments disclosed herein involve devices and methods for attaching FO sensors to structures. Embodiments described herein may be used as protection of FO sensors from the environment. In some embodiments, the disclosed devices facilitate mounting FO sensors to the structures in such a way that strain from the structures is transmitted to the sensors. In some embodiments, the disclosed devices can be used to attach FO sensors to structures in such a way that the sensors are isolated from strain in the structures. The approaches discussed herein provide for attachment of FO sensors that is flexible enough to attach the FO sensors to a variety of different substrates, e.g. concrete, metal, masonry, and wood. The described devices and methods enhance repeatability of the attachment so that the optical fibers may be attached in the same shape and/or length and/or so that at least some or most of the FO sensors have the same pre-strain once attached. The disclosed attachment approaches can be simple and rapid to perform to facilitate the deployment of multiple FO sensors on a structure. The use of an adhesive attachment approach as described below obviates the need to drill holes in the structure or weld to the structure.

[0036] FIG. **1A** shows a top perspective view of an optical fiber mold device **100** and FIGS. **1B** and **1C** show examples of a bottom view of the optical fiber mold device **100** in accordance with some embodiments. FIGS. **1D** and **1E** show an end cross sectional view of the mold device **100**. The mold device **100** includes a first portion **101** comprising a base layer **110** having a longitudinal feature **111** configured to receive an optical fiber **190** (not shown in FIGS. **1A-1C**-but illustrated in cross section in FIGS. **1D** and **1E**). The longitudinal feature **111** may comprise an opening, a groove, and/or a channel in the base layer **110**, for example. In some embodiments, the longitudinal feature **111** may run the entire length of the base layer **110**. The longitudinal feature **111** may run substantially parallel and/or proximate to the central longitudinal axis **191** of the device **100**. Note that the base layer **110** need not be in the form of a polygon as shown in FIG. **1A** through **1E**. The base layer may have one or more curved sides as in FIG. **16**, for example.

[0037] The mold device **100** includes at least one second portion **102** disposed proximate the base layer **110**. The second portion **102** comprises a center wall **120** and front **128** and back **129** end walls. A mold cavity **103** can be configured so that an optical fiber sensor such as a fiber Bragg grating (FBG) sensor can be housed within the mold cavity **103**. The mold device **100** may be formed as a unitary (single-piece) structure.

[0038] There is at least one first hole **121** in the mold cavity **103** is configured to allow mold material, e.g., epoxy, to enter the mold cavity **103**. For example, the first hole **121** may be positioned at the top of the center wall **120** about midway between the front end wall **128** and the back end wall **129** in some configurations. The mold device **100** may include a plug **121p** that blocks the hole **121** after filling the mold cavity **103** so that the mold material, e.g., epoxy does not substantially leak out after injection of the mold material.

[0039] There is at least one second hole **122** into the mold cavity **103** configured to allow air displaced by the mold material to exit the mold cavity **103**. The mold device **100** may include a plug **122p** that can be used to block the hole **122** after filling the mold cavity **103** so that the mold material does not substantially leak out of hole **122** after injection of the mold material. According

to some implementations, the at least one second hole **122** may be positioned in the front end wall **128** and/or the back end wall **129**, e.g., near the top of the front end wall **128** and/or back end wall **129**. The second hole **122** (or multiple holes) can be disposed at defined location(s) that allow homogeneous filling of the mold cavity **103** with few voids. The design of the mold cavity **103** facilitates a defined “laminar” flow of the mold material. After the mold material cures, the optical fiber and/or FO sensor is securely held within the mold cavity **103**.

[0040] The mold device **100** provides a permanent housing for the optical fiber and/or FO sensor. When the mold material enters the mold cavity it comes into contact with the optical fiber and/or FO sensor within the mold cavity. After injection of the mold material into the mold cavity of the mold device and subsequent curing of the mold material, the mold device and mold material seals and protects the fiber and/or FO sensor from environmental stress such as humidity, ultraviolet (UV) radiation, and/or animal intrusion, for example.

[0041] As illustrated in FIG. **1A**, the front end wall **128** and/or the back end wall **129** of the mold cavity may be substantially flat. As discussed below, in some implementations, the front end wall and/or the back end wall of the mold cavity may be curved. FIG. **1A** depicts a mold cavity **103** in which the center wall **120** is curved, e.g., forming a half cylinder. As discussed below, the center wall may have curved or substantially flat sides. In some embodiments, the center wall may have a top that is substantially flat.

[0042] Optionally, the base layer of the mold device includes side, front, and/or back extensions. In some configurations, as depicted in the top perspective view of FIG. **1A**, the base layer **110** includes one or more extensions **131**, **132**, **141**, **142** extending away from the mold cavity walls **120**, **128**, **129**. As illustrated, the side extensions **131**, **132** extend away from the center wall **120**, the front extension **141** extends away from the front end wall **128**, and the back extension **142** extends away from the back end wall **129**.

[0043] As best seen in the bottom view of FIG. **1B** and **1C** and the cross sectional views of FIGS. **1D** and **1E**, the base layer **110** has a first major surface **110a** and an opposing second major surface **110b** with the mold cavity **103** disposed between the first major surface **110a** and the second major surface **110b**. FIG. **1C** shows an example in which the first major surface **110a** and the **110b** are partially within the mold cavity **127** and a portion **125** of the mold cavity **127** is open.

[0044] In embodiments that include extensions, the each side extension **131**, **132** has a first surface **131a**, **132a** and an opposing second surface **131b**, **132b**. The front extension **141** has a first surface **141a** and an opposing second surface **141b**. The back extension **142** has a first surface **142a** and an opposing second surface **142b**. As discussed in more detail below, any or all of these surfaces of the mold device **100** can be suitable for application of an adhesive that attaches the mold device to a structure. Adhesive may be disposed on one or more of the second major surface **110b** of the base layer, the first surfaces **131a**, **132a** of the side extensions, the second surfaces of the side extensions **131b**, **132b**, the first surface of the front extension **141a**, the second surface of the front extension **141b**, the first surface of the back extension **142a**, and the second surface of the back extension **142b**.

[0045] As indicated in FIG. **1B**, one or more of the bonding surfaces **110b**, **131a**, **131b**, **132a**, **132b**, **141a**, **141b**, **142a**, **142b** of the mold device **100** may include capillary channels **155** configured to enhance bonding of the adhesive to facilitate attachment to the structure surface. The capillary channels **155** can also enhance strain transfer from the structure to the optical fiber and/or optical fiber sensor.

[0046] Suitable materials useful for forming the first and/or second portions **101**, **102** of the mold device **100** include plastic, metal, silicone, and/or urethane, for example. The material of the first and/or second portions **101**, **102** of the mold device **100** may be rigid or flexible allowing the mold device **100** to be installed on surfaces that are substantially flat or are not flat. The material of the first **101** and/or second portions may be transparent to wavelengths of visible light and/or to wavelengths of curing radiation. The adhesive used to bond the mold device **100** to the structure

surface may be a fast curing adhesive to allow quick attachment of the optical fiber and/or optical fiber sensor to a structure. Suitable materials for the adhesive include double sided adhesive tape, single sided adhesive tape, a laminated adhesive superglue (cyanoacrylate), and/or epoxy, etc. [0047] Referring now to FIGS. 1D and 1E, the width of the center wall, W1, may be in a range of about 2-10 mm, e.g., about 6 mm; the width of the side extensions, W2, may be in a range of about 2-10 mm, e.g., about 5 mm; and the radius of curvature of the center wall, R, may be in a range of about 1-5 mm, e.g., about 3 mm. The thickness of the mold material may be in a range of about 0.25 mm to about 0.75 mm, e.g., about 0.5 mm. These dimensional ranges are also applicable to other embodiments.

[0048] Turning now to FIG. 2, there is shown a perspective view of a mold device 200 in accordance with some embodiments.

[0049] The mold device 200 includes a first portion 201 comprising a base layer 210 having side extensions 231, 232. The base layer 210 has a first major surface 210a within the mold cavity 203 and an opposing second major surface 210b. The side extensions 231, 232 each have a first major surface 231a, 232a and an opposing second major surface 231b, 232b. As presented in greater detail below, an adhesive may be disposed on the surfaces 210b, 231a, 231b, 232a, 232b to adhere the mold device 200 to a structure.

[0050] The base layer 210 includes a longitudinal feature 211 configured to receive an optical fiber 190. The longitudinal feature 211 may comprise longitudinal opening, channel, or groove running longitudinally along the base layer 210. As shown in the embodiment of FIG. 2, the longitudinal feature 211 may comprise a groove that runs the entire length, L1, of the base layer 210, for example.

[0051] The second portion 202 of the mold device 200 includes a center wall 220 with front and back end walls 228, 229 that are substantially flat. As depicted in FIG. 2, the sides of the center wall 220 may be substantially flat and the top of the center wall 220 may be curved. Alternatively, the top of the center wall may also be substantially flat. The second portion 202 covers a section of the base layer 210 and at least partially covers the longitudinal feature 211 of the base layer 210.

[0052] The first and second portions 201, 202 form a mold cavity 203. There is a first hole 221 in the mold cavity 203 configured to receive mold material into the mold cavity 203. In many configurations, the first hole 221 is located near the top of the center wall 220. The hole 221 may be positioned at any convenient location in the second portion 102 that facilitates injection of the mold material, e.g., about midway along the length of the center wall and about midway along the width of the center wall 220 as shown. There is a second hole 222 into the mold cavity 203 that allows air displaced by the mold material to exit the mold cavity 203. In some cases, there may be multiple holes. For example, a first hole may be disposed on wall 228 and a second hole may be disposed on wall 229. The hole 222 may be positioned at any convenient location in the second portion 102 that facilitates allowing air to escape during injection of the mold material. For example, in some embodiments the second hole 222 may be located near the center top of the front and/or back end walls 228, 229.

[0053] The length of the mold device 200 may be in a range of about 1 cm to about 4 cm, e.g., about 2.5 cm. The diameter of the first hole 221 may be in a range of about 1 mm to 5 mm, e.g., about 3 mm. The diameter of the second hole 222 may be in a range of about 0.5 mm to about 1.0 mm, e.g., about 0.8 mm. The mold device 300 shown in the embodiment of FIG. 3 is similar in some aspects to that of mold device 200 shown in FIGS. 2A and 2B. However, mold device 300 includes a base layer 310 having optional front and back extensions 341, 342 that extend respectively from the front and back end walls 228, 229. The front extension 341 includes a first major surface 341a and an opposing second major surface 341b. The back extension 342 includes a first major surface 342a and an opposing second major surface 342b. As discussed further herein, an adhesive may be applied to one or more of the first major surfaces 341a, 342a and the second major surfaces 341b, 342b to adhesively attach the mold device 300 to a structure.

[0054] The front and back extensions **341**, **342** may include features **351**, **352** that extend at least partially through the extensions **341**, **342** of the base layer **310**. The indentations are suitable to contain glue that secures the optical fiber to the mold device **300** and/or to the structure. For example, in some embodiments, the indentations **351**, **352** may comprise wells that do not extend through both the first major surfaces **341a**, **342a** and the second major surfaces **341b**, **342b**. Alternatively, the indentations may be holes that extend through both the first major surfaces **341a**, **342a** and the second major surfaces **341b**, **342b**. The indentations **351**, **352** may be used to secure the optical fiber **190** for pre-tensioning during installation of the optical fiber **190** and mold device **300** as discussed below.

[0055] FIG. **4** is a top view of a mold device **400** in accordance with some embodiments. The mold device **400** includes a first portion **401** and a second portion **402**. The first portion comprises a base layer **410** having side extensions **431**, **432** and front and back extensions **441**, **442**. The second portion **402** comprises a center wall **420**, front wall **428** and back wall **429**. The side extensions **431**, **432** of the base layer **410** extend from the center wall **420**, the front extension **441** of the base layer **410** extends from the front wall **428**, and the back extension **442** of the base layer **410** extends from the back wall **429**. Note that in this particular embodiment, the center wall, front wall, and back wall **420**, **428**, **429** are all curved.

[0056] A first hole **421** through the center wall **420** is located near the top of the center wall **420** and is configured to receive mold material. A second hole **422** is located in the back end wall **429** near the top of the back end wall. In many implementations, the position of the second hole **422** may be lower than the position of the first hole **451** along the z-axis. In some cases, the second hole **422** may include multiple second holes. For example, a hole may be disposed on wall **428** and a hole may be disposed on wall **429**.

[0057] A longitudinal feature **411** comprising a longitudinal opening, groove, and/or channel that extends longitudinally in the base layer **410** along the y-axis. In this embodiment, the front extension **428** includes a hole **451** that extends all the way through the front extension layer **428**. The back extension **429** includes a hole **452** that extends all the way through the back extension layer **429**. Note that in this embodiment, the holes **451**, **452** in the front and back extensions **428**, **429** merge with the longitudinal opening **411**. According to various embodiments, hole **451** and/or hole **452** may comprise a well that includes a hole having walls.

[0058] FIGS. **5A** and **5B** are directed to a mold device **500-1** in accordance with some embodiments. FIG. **5A** is a side cross sectional view of the mold device **500-1** and structure **199**. FIG. **5B** shows a bottom view of the mold device **500-1**. Mold device **500-1** includes first portion **501-1** comprising a base layer **510-1** without side, front, or back extensions. A longitudinal feature **511** is configured to receive an optical fiber **190**. A second portion **502-1** is disposed on the base layer **510-1** such that the second portion **502-1** and the base layer **510-1** form a mold cavity **503-1**. As depicted in FIGS. **5A** and **5B**, the base layer **510-1** has a first major surface **510-1a** proximate the mold cavity **503-1** and an opposing second major surface **510-1b**. An adhesive **560**, e.g., a glue, such as epoxy, super glue (cyanoacrylate) and/or double sided adhesive tape, is disposed on the second major surface **510-1b** of the base layer **510-1** on either side of the mold cavity **503-1**. When the mold device **500-1** is installed on a structure **199**, the adhesive **560** is located between the surface of the structure **199** and the second surface **510-1b** of the base layer **510-1**. The adhesive **560** is selected to form a bond between the mold device **500-1** and the structure **199**.

[0059] FIG. **5C** is a side cross sectional view of a mold device **500-2** and structure **199** in accordance with some embodiments. FIG. **5D** shows a bottom view of the mold device **500-2**. Mold device **500-2** includes a first portion **501-2** comprising a base layer **510-2** having side extensions **531**, **532**. A longitudinal feature **511** is configured to receive and secure an optical fiber **190**. A second portion **502-2** is disposed on the base layer **510-2** such that the second portion **502-2** and the base layer **510-2** form a mold cavity **503-2**. The second portion **502-2** includes a first hole **521** as shown. As best seen in the bottom view of FIG. **5D**, side extension **531** has a second major

surface **531b** and side extension **532** has a second major surface **532b**. In the illustrated embodiment, an adhesive **560**, e.g., glue, such as epoxy, superglue (cyanoacrylate), or double sided adhesive tape, is disposed on the second major surfaces **531b**, **532b** of extensions **531**, **532**. When the mold device **500-2** is installed on a structure **199**, the adhesive **560** is located between the surface of the structure **199** and the second surfaces **531b**, **532b** of the extensions **531**, **532**. The adhesive **560** is selected to form a bond between the mold device **500-2** and the structure **199**. [0060] FIGS. **6A** through **6C** illustrate an embodiment in which adhesive tape **660** e.g., single sided adhesive tape, is disposed on the first surfaces **631a**, **632a** of the side extensions **631**, **632** of a mold device **600** in accordance with some embodiments. The adhesive tape **660** extends across the first surfaces **631a**, **632a** of the side extensions and extends over the surface of the structure **199** thus adhering the mold device **600** to the structure **199**. FIG. **6A** shows a side cross sectional view of the mold device **600**. FIG. **6B** shows the bottom view of the mold device **600** and FIG. **6C** provides a perspective view of the mold device **600** and structure **199**. Mold device **600** includes a first portion **601** comprising a base layer **610** having side extensions **631**, **632**. A second portion **602** is disposed on the base layer **610** such that together the second portion **602** and the base layer **610** form a mold cavity **603** as discussed previously. The second portion **620** includes a first hole **621** in the center wall as shown. As best seen in the bottom view of FIG. **6B**, side extension **631** has a second major surface **631b** and side extension **632** has a second major surface **632b**. The second major surfaces **631b**, **632b** make contact with the surface of the structure. A longitudinal feature **611** is configured to receive and secure an optical fiber **190**.

[0061] As best seen in the perspective view of FIG. **6C**, side extension **631** has a first major surface **631a** and side extension **632** has a first major surface **632a**. In the illustrated embodiment, an adhesive **660**, which may be a single sided tape, is disposed on the first major surfaces **631a**, **632a** of extensions **631**, **632** and extends beyond the extensions **631**, **632** onto the surface of the structure **199**. The adhesive **660** is selected to form a bond between the surface of the structure **199** and the mold device **600** so as to securely hold the mold device **600** against the structure **199**.

[0062] FIG. **7** illustrates a mold device **700** in accordance with some embodiments. In this particular embodiment, a flowable glue, such as superglue (cyanoacrylate), or epoxy, is used to bond the mold device **700** to the structure **199**. The glue **760** is disposed on the top surfaces **731a**, **732a** of the side extensions **731**, **732** of the mold device **700** and extends out over a portion of the surface of the structure **199**.

[0063] FIGS. **8A** and **8B** show a mold device **800** in accordance with embodiments. FIGS. **8A** and **8B** respectively show a bottom view and side cross sectional view of device **800**. The mold device **800** shown in FIGS. **8A** and **8B** has a number of features that are substantially similar to the features of mold device **500-2** previously discussed and illustrated in FIGS. **5C** and **5D**. In FIGS. **8A** and **8B**, the same reference numbers are used to refer to the substantially similar features shown in FIGS. **5C** and **5D**. FIG. **8A** depicts a bottom view of mold device **800**. FIG. **8B** is a side cross sectional view of mold device **800**.

[0064] The mold device **800** differs from device **500-2** in that mold device **800** includes front and back extensions **841**, **842** extending away from the front and back walls **528**, **529**, respectively. An adhesive **860** is disposed on the front and back extensions **841**, **842**. Each of the front and back extensions **841**, **842** respectively include front and back features **851**, **852** in the extension **531**, **532**. In this particular embodiment, features **851**, **852** are holes through the extension **531**, **532**. As depicted in the cross sectional diagram of FIG. **8B**, hole **852** includes a lip **852a**. Optionally one or both holes include a lip **852a** that is configured to contain glue within the hole after the glue is deposited in the hole **852**.

[0065] As best seen in the cross sectional diagram of FIG. **8B**, each extension **841**, **842** includes a first major surface **841a**, **842a** proximate the front and back walls **528**, **529** and an opposing second surface **841b**, **842b**. An adhesive **860** is disposed on the second major surfaces **531b**, **532b** of the side extensions **531**, **532** such that the adhesive **860** extends over the surfaces and **531b**, **532b** of

the side extensions **531**, **532** and over the second major surfaces **841b**, **842b** of the front and back extensions **841**, **842**. Additional adhesive is shown on the second surface **842b** of back extension **842**.

[0066] The mold device **900** shown in FIGS. **9A** and **9B** has a number of features that are substantially similar to the features of mold device **600** depicted in FIGS. **6A** through **6C**. In [0067] FIGS. **9A** and **9B**, the same reference numbers are used to refer to the substantially similar features shown in FIGS. **6A** through **6C**. FIG. **9A** depicts a bottom view of mold device **900**. FIG. **9B** is a perspective view of mold device **900**.

[0068] The mold device **900** differs from device **600** in that mold device **900** includes front and back extensions **941**, **942** extending away from the front and back walls **628**, **629**, respectively. Each of the front and back extensions **941**, **942** includes a feature **951**, e.g., a well or hole **952** in the extension layer.

[0069] As best seen in the diagram of FIG. **9B**, each extension **941**, **942** includes a first major surface **941a**, **942a** proximate the front and back walls **628**, **629** and an opposing second surface **941b**, **942b**. An adhesive **960** is disposed on the first major surfaces **631a**, **632a** of the side extensions **631**, **632** such that the adhesive **960** extends over the surfaces and **631a**, **632a** of the side extensions **631**, **632** and also over the first major surfaces **941a**, **942a** of the front and back extensions **941**, **942**. The adhesive, which may be a single-sided tape, extends from the extensions **631**, **632**, **941**, **942** out over the surface of the structure **199** so as to adhere the mold device **900** to the structure **199**.

[0070] According to some implementations, two or more of the mold devices may be used together to form sensing clusters. This technique is useful for sensing a parameter in multiple dimensions, for example. A particularly useful configuration is the biaxial configuration in which two sensors are arranged along two orthogonal axes (see, e.g., FIG. **12**). Another useful configuration is a rosette configuration in which two sensors are arranged along two orthogonal axes and a third sensor is arranged at an angle of about **45** degrees to the orthogonal axes (see, e.g., FIG. **11A**). When sensors are used in clusters, the shape of mold device can be designed to facilitate alignment of multiple mold devices. The multiple mold devices can be used to bond to multiple fiber sensors in certain well defined configurations. Engagement features on the mold devices may provide for orientation of the mold devices and/or for locking the mold devices together.

[0071] FIG. **10** is a diagram showing a mold device **1000** that includes engagement features useful to orient and/or interlock multiple mold devices in a pattern. Mold device **1000** comprises a first portion **1001** that includes a base layer **1010** having side extensions **1031**, **1032** and front and back extensions **1041**, **1042**. A second portion **1002** is disposed on the base layer **1010**, the second portion **1002** including a center wall **1020** and front and back walls **1028**, **1029**. According to some embodiments, the base layer **1010** forms a polygon having substantially straight edges **1081**, **1082**, **1083**, **1084**, **1085**, **1086**. Some of the edges are oriented at **90** degrees to one another, e.g., edges **1081**, **1082** and some edges are sloped, e.g., oriented at angles different from **90** degrees. For example, edge **1085** is oriented at a **45** degree angle with respect to edge **1081** and edge **1086** is oriented at a **45** degree angle with respect to edge **1083**. When multiple mold devices are used together in a pattern, the sloped edges can be used to make mitered corners, e.g., see FIG. **12**.

[0072] In some embodiments, the edges may include additional engagement features that orient and/or lock the mold devices together. For example, along edges **1081** and **1086**, mold device **1000** includes protrusions **1071** that are compatible to receive indentations **1072** disposed on edges **1083** and **1085** of another mold device. Other engagement features may include sloped sides, grooves and/or other features that serve to orient, connect, and/or lock the mold devices together. According to various embodiments, the configuration shown in FIG. **10** may include one or more of longitudinal openings, grooves, channels and/or temporary epoxy openings similar to features **951** and **952**.

[0073] FIG. **11A** shows a cluster **1100** of mold devices **1100-1**, **1100-2**, **1100-3** arranged in a rosette

pattern. The sloped edge **1186-2** of mold device **1100-2** is arranged so that it is adjacent to straight edge **1183-1** of device **1100-1**. The sloped edge **1185-2** of mold device **1100-2** is arranged adjacent to straight edge **1181-3** of device **1100-3**. The arrangement of edges **1183-1**, **1186-2**, **1185-2**, **1181-3** forms a mitered corner between the mold devices **1100-1**, **1100-2**, **1100-3**.

[0074] FIG. **11B** provides another example of mold devices **1100-1**, **1100-2**, **1100-3** arranged in a rosette pattern with representative dimensions. The length, **L2**, of mold device **1100-1**, **1100-3** is in a range of about 2 to 10 cm, e.g. about 6 cm. The length, **L3**, of mold device **1100-2** is in a range of about 2 to 10 cm, e.g. about 5.7 cm. The width, **W6**, of the mold devices **1100-1**, **1100-2**, **1100-3**, is in a range of about 0.5 cm to about 2.5 cm, e.g., about 1.7 cm. The center-to-center distance, **L4**, between the front feature and the back feature is in a range of about 2 cm to 6 cm, e.g., about 4 cm. The center-to-center distance, **L5**, between the back feature and the first hole in the second portion is in a range of about 1 cm to about 3 cm, e.g., about 2 cm. The distance, **L6**, between the front edge and the center of the front feature is in a range of about 0.5 mm to about 2 mm, e.g., about 1 mm. FIG. **12** shows a cluster **1200** of mold devices **1200-1**, **1200-2** arranged in a biaxial pattern. The sloped edge **1286-2** of mold device **1200-2** is arranged so that it is adjacent to the sloped edge **1285-1** of device **1200-1**. The arrangement edges **1286-2**, **1285-1** forms a mitered corner between the mold devices **1200-1**, **1200-2**.

[0075] FIG. **13A** is a flow diagram illustrating a method of using the mold device to affix an optical fiber to the surface of a structure. The method of FIG. **13A** is suitable when the longitudinal feature includes a longitudinal opening in the base layer. According to this method, the optical fiber is positioned **1305** on the surface of the structure. The mold device is positioned **1315** on the surface of the structure such that the optical fiber is received into the longitudinal opening. The mold device is adhesively attached **1325** to the structure surface by an adhesive disposed on the top and/or bottom surfaces of the mold device. Optionally, the optical fiber is pre-tensioned **1335** before mold material is injected **1345** into the mold device.

[0076] FIG. **13B** is a flow diagram illustrating a method of using the mold device to affix an optical fiber to the surface of a structure. In this particular embodiment, an optical fiber is positioned **1310** in longitudinal feature of a mold device. The mold device and optical fiber are placed **1320** on the surface of the structure. The mold device is adhesively attached **1330** to the structure surface by an adhesive disposed on the top and/or bottom surfaces of the mold device. Optionally, the optical fiber is pre-tensioned **1340** before mold material is injected **1350** into the mold device.

[0077] FIG. **14** illustrates a cluster of mold devices **1401**, **1402**, **1403** holding optical fibers **190-1**, **190-2**, **190-3** in a rosette pattern. The central axes of optical fiber **190-3** and mold device **1403** are arranged to lie along the x-axis; the central axes of optical fiber **190-1** and mold device **1401** lie along the y-axis which is orthogonal to the x-axis ($\theta_{\text{sub.2}}=90$ degrees); and the central axes of optical fiber **190-2** and mold device **1402** are arranged at an angle, $\theta_{\text{sub.1}}=45$ degrees, with respect to the x-axis. In this embodiment, the optical fibers **190-1**, **190-2**, **190-3** are secured in the mold cavities of the mold devices **1401**, **1402**, **1403** by mold material injected through the first hole of each mold device **1401**, **1402**, **1403**. Each of the mold devices **1401**, **1402**, **1403** are secured to the surface of the structure **199** by adhesive disposed on the top and/or bottom surfaces of the devices **1401**, **1402**, **1403**. The optical fibers **190-1**, **190-2**, **190-3** are attached by epoxy or other type glue to the structure at two or more attachment points. Optical fiber **190-1** is attached to the surface of at attachments points **1495-1a**, **1495-1b**; optical fiber **190-2** is attached at attachments points **1495-2a**, **1495-2b**; and optical fiber **190-3** is attached at attachments points **1495-3a**, **1495-3b**. The optical fibers **190-1**, **190-2**, **190-3** may optionally be pre-strained before attachment to the structure **199** at attachment points **1495-1a**, **1495-1b**, **1495-2a**, **1495-2b**, **1495-3a**, **1495-3b**.

[0078] FIGS. **15A** through **15Q** illustrate a method of attaching optical fibers to a structure in a rosette pattern using mold devices in accordance with some embodiments. Referring now to FIG. **15A**, initially, an optical fiber **1591** is positioned on a structure **199**. As shown in FIG. **15B**, the first mold device is placed over the optical fiber **1591**. The longitudinal opening **1511** in the mold

device **1501** receives the optical fiber **1591**. The first mold device **1501** is positioned so that the optical fiber is exposed by the first hole **1551** in the front extension of the first mold device **1501** and by the second hole **1552** in the back extension of the first mold device **1501**. The first mold device **1501** is attached to the surface of the structure **199** by an adhesive disposed on the top and/or bottom surfaces of the mold device **1501**. An additional mold device **1503** is positioned on the surface. The sloped corner of the additional mold device **1503** assists in orienting the second mold device **1503** at about a 45 degree angle with respect to the first mold device **1501**.

[0079] FIG. **15C** shows the arrangement after a flowable glue, such as epoxy **1560** has been deposited into the hole **1551** in the front extension of the first mold device **1501** and the epoxy is cured. Note that the epoxy used in hole **1551** need not be the same adhesive as the adhesive used to secure the mold device to the surface of the structure. The epoxy **1560** bonds one end of the optical fiber **1591** to the surface of the structure **199**. Tension is applied to the optical fiber **1591** in the direction of arrow **1596** as depicted in FIG. **15D**. While the tension is being applied along the direction **1596**, epoxy is deposited into the second hole **1552** in the back extension of the first mold device **1501** and the epoxy is cured as shown in FIG. **15E**. The cured epoxy in the first and second holes **1551**, **1552** maintains the tension in optical fiber **1591** at a predetermined pre-tension value. The tension applied to the optical fiber **1591** along direction **1596** is released as indicated in FIG. **15F**.

[0080] Referring now to FIG. **15G**, in a next process step, optical fiber **1592** is positioned on the structure **199**. The second mold device **1502** is placed over the optical fiber **1592**. The longitudinal opening **1511** in the mold device **1502** receives the optical fiber **1592**. The second mold device **1502** is positioned so that the optical fiber **1592** is exposed by the first and second holes **1551**, **1552** in the second mold device **1502**. The sloped corner of the third mold device **1503** assists in orienting the second mold device **1502** at about a 45 degree angle with respect to the third mold device **1503**.

[0081] FIG. **15H** shows the arrangement after epoxy **1560** has been deposited into the hole **1551** in the front extension of the second mold device **1502** and the epoxy is cured. The epoxy **1560** bonds one end of the optical fiber **1592** to the surface of the structure **199**. Tension is applied to the optical fiber **1592** in the direction of arrow **1597** as depicted in FIG. **15I**. While the tension is being applied along the direction **1597**, epoxy is deposited into the second hole **1552** in the back extension of the second mold device **1502** and the epoxy is cured as shown in FIG. **15J**. The cured epoxy in the first and second hole **1551**, **1552** of the second mold device **1502** maintains the tension in optical fiber **1592** at a predetermined pre-tension value. The tension applied to the optical fiber **1592** along direction **1596** is released.

[0082] In the next step, the third mold device is rotated (FIG. **15K**) out of the plane of the structure surface to allow the third optical fiber **1593** to be positioned on the surface of the structure **199** (FIG. **15L**). The third mold device **1503** is then oppositely rotated onto the surface such that the longitudinal opening **1511** in the third mold device **1503** receives the optical fiber **1593** (FIG. **15M**). The third mold device **1503** is positioned so that the optical fiber **1593** is exposed by the first and second holes **1551**, **1552** in the third mold device **1503**.

[0083] FIG. **15N** shows the arrangement after epoxy **1560** has been deposited into the hole **1551** in the front extension of the third mold device **1503** and the epoxy is cured. The epoxy **1560** bonds one end of the optical fiber **1593** to the surface of the structure **199**. Tension is applied to the optical fiber **1593** in the direction of arrow **1598** as depicted in FIG. **15O**.

[0084] While the tension is being applied along the direction **1598**, epoxy is deposited into the second hole **1552** in the back extension of the third mold device **1503** and the epoxy is cured as shown in FIG. **15P**. The cured epoxy in the first and second holes **1551**, **1552** of the third mold device **1503** maintains the tension in optical fiber **1593** at a predetermined pre-tension value. The tension applied to the optical fiber **1593** along direction **1598** is released. A mold material is injected into the mold cavities of the first, second, and third mold devices **1501**, **1502**, **1503**

through holes **1561**. Note that the mold material may be an epoxy and need not be the same material as used to secure the optical fibers **1591**, **1592**, **1593** and/or to attach the mold devices **1501**, **1502**, **1503** to the structure **199**.

[0085] The embodiments previously described are suitable for securing optical fibers that include fiber optic sensors, e.g., fiber Bragg grating (FBG) sensors. FBG sensors can be sensitive to strain in the structure. The approaches discussed herein can facilitate mechanically coupling an FBG sensor to a structure so that strain in the structure is transferred to the sensor.

[0086] Conversely, some implementations involve bonding an optical fiber comprising an FBG sensor in such a way that the strain in the structure is not substantially transferred to the sensor. An FBG sensor that is substantially strain-isolated can be used to measure temperature near the structure, for example.

[0087] FIG. **16** is a top view of a mold device **1600** suitable for attaching an optical fiber comprising a FBG temperature sensor to a structure such that the temperature sensor is substantially strain-isolated from the structure **199**. The mold device **1600** includes a first portion **1601** comprising a base layer **1610** having two curved sides **1681**, **1683** and two straight sides **1682**, **1684** in this implementation. A longitudinal feature **1611** such as a longitudinal opening or longitudinal groove runs along the base layer **1610**.

[0088] Two second portions **1602-1**, **1602-2** are disposed over the first portion **1601**. Each second portion **1602-1**, **1601-2** comprises a center wall **1620**, a front end wall **1628**, and an end wall **1629**. The base layer **1610** and the second portion **1602-1** form a first mold cavity **1603-1**. The base layer **1610** and the second portion **1602-2** form a second mold cavity **1603-2**. First holes **1621** through the center walls **1620** of the second portions **1602-1**, **1602-2** are configured to allow mold material to be injected into the mold cavities **1603-1**, **1603-2**.

[0089] Second holes (not shown) are disposed in the second portions **1602-1**, **1602-2** to allow air displaced by the mold material to exit the mold cavities **1603-1**, **1603-2**.

[0090] A tunnel portion **1604** is disposed between the first and second portions **1602-1**, **1602-2**. The tunnel portion **1604** is configured to receive and at least partially enclose a fiber optic sensor so that the sensor is substantially mechanically isolated from strain in the tunnel portion **1604**. In some embodiments, the tunnel portion **1604** does not include a hole for mold injections and/or air exit. The relatively small defined tunnel portion **1604** housing the optical fiber temperature sensor serves to reduce the temperature sensor exposure to pre-strain and reduces the likelihood that the temperature sensor is accidentally “bonded” to the structure by stain, coatings, paint, etc.

[0091] The base layer **1610** includes side extensions **1631**, **1632** that extend away from the center walls **1620** of the second portions **1602-1**, **1602-2**. A front extension **1641** extends away from the front wall **1628** of the first portion **1602-1**. A hole **1651** configured to receive epoxy is located in the front extension **1641**. A back extension **1642** extends away from the back wall **1629** of the second portion **1602-2**. A hole **1652** configured to receive epoxy is located in the back extension **1642**.

[0092] The base layer **1610** has a first major surface proximate the second portions **1602-1**, **1602-2** inside the mold cavity **1603-1**, **1603-2** and an opposing second major surface. Side extensions **1631**, **1632** have first major surfaces proximate the second portions **1602-1**, **1602-2** and opposing second surfaces **1631b**, **1632b**. Front and back extensions **1641**, **1642** have first major surfaces proximate the second portions **1602-1**, **1602-2** and opposing second surfaces. As previously discussed, adhesive may be disposed on one or more of the surfaces to adhesively bond the mold device **1600** to the structure surface as previously discussed.

[0093] FIG. **17A** is a top view and FIG. **17B** is a side view of another mold device **1700** suitable for attaching an optical fiber comprising a strain free temperature sensor and/or a span sensor. In this example, the sensor may not be directly attached to structures, but is disposed and protected in the enclosed mold structure. In some cases, two corners of the sensor is attached to the mold instead of directly attached to the structure.

[0094] The mold device **1700** includes a first portion **1701** comprising a base layer **1710** having a first two opposing substantially straight sides **1781**, **1783** and a second two opposing substantially straight sides **1782**, **1784**. A longitudinal feature **1711** such as a longitudinal opening or longitudinal groove runs along the base layer **1710**.

[0095] Two second portions **1702-1**, **1702-2** are disposed over the first portion **1701**. Each second portion **1702-1**, **1701-2** comprises a center wall **1720**, a front end wall **1728**, and an end wall **1729**. The base layer **1710** and the second portion **1702-1** form a first mold cavity **1703-1**. The base layer **1710** and the second portion **1702-2** form a second mold cavity **1703-2**. First holes **1721** through the center walls **1720** of the second portions **1702-1**, **1702-2** are configured to allow mold material to be injected into the mold cavities **1703-1**, **1703-2**. Second holes (not shown) are disposed in the second portions **1702-1**, **1702-2** to allow air displaced by the mold material to exit the mold cavities **1703-1**, **1703-2**.

[0096] A tunnel portion **1704** is disposed between the first and second portions **1702-1**, **1702-2**. The tunnel portion **1704** is configured to receive and at least partially enclose a fiber optic sensor so that the sensor is substantially protected from environmental perturbation (e.g., wind, rain, etc.) in the tunnel portion **1704**. According to various configurations, the temperature sensor is substantially mechanically isolated from strain in the tunnel portion **1704**, and the span sensor is pre-strained in the tunnel portion **1704**. In some embodiments, the tunnel portion **1704** does not include a hole for mold injections and/or air exit. The relatively small defined tunnel portion **1704** housing the optical fiber temperature sensor serves to reduce the temperature sensor exposure to pre-strain and reduces the likelihood that the temperature sensor and/or span sensor is accidentally “bonded” to the structure by stain, coatings, paint, etc.

[0097] The base layer **1710** includes side extensions **1731**, **1732** that extend away from the center walls **1720** of the second portions **1702-1**, **1702-2**. A front extension **1741** extends away from the front wall **1728** of the first portion **1702-1**. A hole **1751** configured to receive epoxy is located in the front extension **1741**. A back extension **1742** extends away from the back wall **1729** of the second portion **1702-2**. A hole **1752** configured to receive epoxy is located in the back extension **1742**.

[0098] The base layer **1710** has a first major surface proximate the second portions **1702-1**, **1702-2** inside the mold cavity **1703-1**, **1703-2** and an opposing second major surface. Side extensions **1731**, **1732** have first major surfaces proximate the second portions **1702-1**, **1702-2** and opposing second surfaces. Front and back extensions **1741**, **1742** have first major surfaces proximate the second portions **1702-1**, **1702-2** and opposing second surfaces. As previously discussed, adhesive may be disposed on one or more of the surfaces to adhesively bond the mold device **1700** to the structure surface as previously discussed.

[0099] The mold devices described herein facilitate defining the shape, size, location, and/or amount of adhesive used for the bond between the mold device and the structure. The disclosed approaches for using the mold device enable a defined and homogeneous strain transfer (within one sensor or between different sensors), e.g., by reducing bubbles and/or voids in the mold material. The mold device enables controlled bonding which can be implemented into automated, semi-automated, and/or manual fiber sensor installation using robots and/or installation tools. When installed as discussed herein, the mold device protects the optical fiber and/or fiber sensor from environmental impacts such as water ingress, UV exposure, chemical impact such as car exhaust, humidity, vibration, etc.

[0100] In some embodiments, the type of adhesive and/or mold material used with the mold device may be configured to indicate that bonding occurs and/or was performed correctly. For example, with reference to FIG. **8B**, a first type adhesive may be used to bond the mold device **800** to the structure surface, a second type of adhesive may be used in features **851**, **852**, and a third type of adhesive may be injected into hole **521** as the mold material. In some embodiments, different types of adhesive may change color when deposited as an indicator to facilitate installation. Adhesive

materials used at different locations may be different types. For example, adhesive used to bond the mold device to the structure may be a fast curing adhesive. The adhesive deposited in features **851**, **852** and/or **521** may be a slower curing adhesive, for example.

[0101] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as representative forms of implementing the claims.

Claims

1-14. (canceled)

15. A method, comprising: positioning an optical fiber in a longitudinal feature of a planar base layer of a mold device, the mold device comprising: a first portion comprising the planar base layer having the longitudinal feature along a plane of the planar base layer, the longitudinal feature configured to receive the optical fiber, the planar base layer comprising peripheral planar portions adjacent to the longitudinal feature; and a second portion disposed over the peripheral planar portion, the second portion comprising: a center wall; and front and back end walls, the center wall, the front end wall, and the back end wall forming a mold cavity; at least one first hole in the mold cavity configured to allow mold material to enter the mold cavity; and at least one second hole in the mold cavity configured to allow air displaced by the mold material to exit the mold cavity; placing the mold device and the optical fiber on a structure surface; and injecting the mold material into the first hole of the mold device.

16. The method of claim 15, further comprising adhesively attaching the mold device to the structure surface.

17. The method of claim 15, wherein: the base layer comprises first and second side extensions that extend in opposite directions away from the center wall; and the method further comprises adhesively attaching the mold device to the structure surface at the first and second side extensions.

18. The method of claim 15, further comprising pre-straining the optical fiber before injecting the mold material.

19. The method of claim 15, wherein the base layer further comprises; a front extension adjacent the front end wall and extending away from the front end wall; a back extension adjacent the back end wall and extending away from the back end wall, the longitudinal feature continuing longitudinally through the front extension and the back extension; a front feature in the front extension aligned with a longitudinal opening and extending through opposing major surfaces of the front extension; a back feature in the back extension aligned with the longitudinal opening and extending through opposing major surfaces of the back extension; and wherein the method further comprises applying glue into the first hole and the second hole and in contact with the optical fiber.

20. The method of claim 19, wherein applying the glue into the front feature and the back feature and in contact with the optical fiber comprises: applying the glue into the back feature; pre-straining the optical fiber after applying the glue into the back feature; and applying the glue into the front feature after pre-straining the optical fiber, wherein after curing, the glue in the front feature and the back feature is configured to hold the optical fiber at a predetermined pre-strain value.

21. The method of claim 15, wherein the longitudinal feature comprises a longitudinal opening.

22. The method of claim 15, wherein the longitudinal feature comprises a longitudinal groove.

23. The method of claim 15, wherein the base layer comprises first and second side extensions that extend in opposite directions away from the center wall.

24. The method of claim 23, wherein an adhesive is disposed on a surface of one or both of the first and second side extensions.

25. The method of claim 15, further comprising a partial base layer, wherein the front and back end

walls, the center wall, the front end wall, the back end wall and the partial base layer form the mold cavity.

26. The method of claim 15, further comprising: a front feature in a front extension aligned with the longitudinal feature and configured to receive adhesive that secures the optical fiber; and a back feature in the back extension aligned with the longitudinal feature and configured to receive adhesive to secure the optical fiber.

27. The method of claim 26, wherein one or both of the front and back features comprises a hole through the base layer.

28. The method of claim 26, wherein one or both of the front and back features comprises a well.

29. The method of claim 15, wherein the well has a lip that extends above the base layer.

30. The method of claim 15, further comprising engagement features disposed on one or more of edges of the base layer, each of the engagement features configured to engage a compatible engagement feature of another optical fiber mold device.
